In-Flight Reconfiguration of a Cable-Suspended Ring Carried by a Team of Quadrotors

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***Abstract*—Teams of quadrotors carrying a cable-suspended payload usually keep a fixed composition. We show that model predictive control lets such a team drop and re-add carriers in flight. A planner solves an optimal control problem over a 0.86 kg ring; because the attachment geometry is a runtime parameter, a change of team is a switch to a pre-built problem of the new size, warm-started from the measured state. Three reconfigurations are handled: an unannounced release, detected from motion capture; a departed carrier's rejoin; and the attach of a newcomer, admitted after magnet contact and a motion-capture proof of co-motion. On a motion-capture rig an unannounced release is detected in 0.42 s and the ring's tilt peaks at 6.5°, back within its band 1.8 s later. In a digital twin of the rig, the tilt after a release grows steeply with the largest gap the survivors leave, a 165° gap crossing the 30° drop line on 1 m cables, and feeding the trackers the measured cable pull rather than the planned one cuts the tilt rise to about two thirds. In the twin, folding a newcomer in at its measured catch point tolerates catch-point errors up to 16 cm radially and 6 cm along the rim, and a departed carrier rejoins its own plate in 10 of 11 runs within a 5° tilt bar. Offline, the solver iterations needed per tick grow with fleet size, from two up to four carriers to five at eight.**

***Index Terms*—Aerial systems: mechanics and control, multi-robot systems, cooperative manipulation, model predictive control, cable-suspended payload.**

I. Introduction

Several quadrotors sharing a payload on cables can lift what none of them could carry alone, and spread the load so that each flies with margin [1], [2]. Such a team is almost always treated as fixed: the carriers are attached on the ground, take off together and land together. A team that could change in the air would be more useful and more robust. A carrier with a low battery or a fault could leave and come back; a fresh carrier could join to take up a heavier load or relieve another; and a carrier that is lost without warning, a released magnet or a failed motor, need not bring the payload down with it. This letter asks what a controller needs to let a team carrying a cable-suspended payload lose and gain carriers in flight.

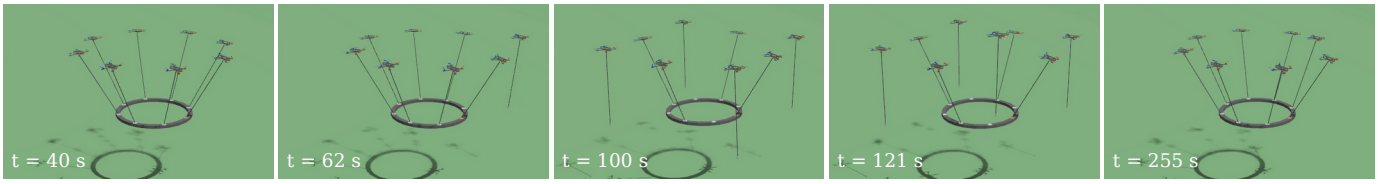
Cooperative cable transport. The statics of a rigid payload hung from several cables, and the internal forces the team may choose freely, are well understood [3]–[5]. Controllers range from decentralised schemes that need no communication between carriers [4], [6], [7] through geometric and leader–follower designs [8], [9]. Model predictive control (MPC) suits the problem because cable tensions are bounded and the payload's motion couples every carrier: nonlinear MPC of the full payload pose [10], [11], kinodynamic planning for the team [12], and distributed formulations [13], [14] all fly a team of a fixed size.

Reconfiguration and attach. Plug-and-play MPC adds and removes subsystems from a distributed controller [15]–[18], but for systems of loosely coupled plants, not for carriers that share one payload. Modular aerial robots dock and undock in flight [19]–[21], and magnetic grippers let a multirotor pick up and release objects [22], [23]. Fault-tolerant cooperative transport keeps a payload up after a carrier fails [24], [25]. We know of no work in which a team carrying a cable-suspended payload both loses carriers without warning and admits carriers that join in the air, nor of one flown on hardware.

The working hypothesis of this letter is that a centralised optimal control problem suits a change of team because the planner solves for the force distribution of the team it actually has in the first tick after the event, instead of leaving the survivors to find it from the motion that follows. The contributions are:

- a reconfiguration method: a cooperative MPC whose attachment geometry is a runtime parameter, so that a change of team is a switch to a pre-built problem of the new size, warm-started from the measured state; with it, motion-capture detection of an unannounced release, a drop rule from the survivors' largest gap, the rejoin of a departed carrier, and the admission of a newcomer on a motion-capture proof of co-motion (Sections III, III-B and III-C);
- what governs the release transient: the ring's tilt after a release grows steeply with the largest gap the survivors leave, so that on 1 m cables a 165° gap crosses the 30° drop line, and feeding the trackers the measured cable pull rather than the planned one cuts the tilt rise (Section IV-B);
- the fold-in's tolerance to error in the newcomer's catch point (twin): up to 16 cm radially when the measured catch point is used against 6 cm when the nominal plate point is assumed, and 6 cm along the rim (Section IV-C);
- validation of the release on a motion-capture rig with unmodified Betaflight quadrotors, and of release, rejoin and attach in a digital twin built from the rig's identified thrust map and geometry, including an exploratory chain in which eight carriers drop to four and rejoin all four; and the scaling of the solver's iterations with fleet size (Sections IV-A and IV-D).

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(a) 8 carriers (b) release (c) 4 carriers (d) rejoin (e) 8 again

Fig. 1. In-flight reconfiguration in the digital twin (Gazebo), one run. (a) Eight quadrotors carry the ring on 1 m cables. (b) Carriers release one at a time and fly clear with their cables; two have left. (c) Four carriers hold the ring after four releases. (d) A released quadrotor rejoins, just before its cable tip reaches a free plate. (e) All four have rejoined and eight carriers hold the ring again. Times are from the start of the run.

II. System

Payload. The payload is a ring of mass $m_L = 0.86\,\mathrm{kg}$ carrying twelve magnet plates every 30° at radius $r = 0.225\,\mathrm{m}$ (Fig. 2). Plates are numbered clockwise from the ring's $+x$ axis seen from above. A carrier hangs from a plate on a link of length l_i (0.55 m rods in the rig flights reported here, 1 m cables in the twin unless stated), whose lower end holds an electromagnet switched over the carrier's radio channel; the links fly at about 60° from horizontal. The magnet catches only on contact with its plate, so a release needs no mechanism beyond switching it off.

Rig. The carriers are 0.55 kg quadrotors on unmodified Betaflight flight controllers, whose inner rate loop is kept; our trackers send a collective throttle and three body-rate commands over the radio link. A motion-capture system measures the pose of every quadrotor, the ring and the links' lower ends in a $6 \times 6 \times 4\,\mathrm{m}$ cage, and a position limit of $\pm 1.75\,\mathrm{m}$ lands the fleet if any quadrotor leaves it. Thrust is affine in throttle, with an offset that grows with the carried mass and falls with battery voltage; the map was identified on the rig from free hovers with hung masses over the battery's range.

Digital twin. A Gazebo [26] model under ROS 2 [27] reproduces the rig: the same masses, geometry, identified thrust map and rate-loop model, the motion-capture velocity filter, and the same planner, trackers and safety logic, launched unchanged against simulated topics. The twin models every link as rigid, whereas the rig's tethers can go slack; slack-cable effects and the rig's unmodelled aerodynamics are therefore its honest limit, and the rig flights are the check on it.

III. Method

The controller is a two-level cascade. A centralised planner plans the motion of the ring and of every link over a horizon; a tracking MPC on each carrier follows that carrier's share of the plan. A change of team is handled inside the planner, by changing the problem it solves.

A. Cooperative planner and trackers

The planner's model is the ring as a rigid body, pose $(\mathbf{p}_L, \mathbf{q}_L)$ and twist $(\mathbf{v}_L, \boldsymbol{\omega}_L)$, with each of the N carriers reduced to a link of length l_i attached at the plate point $\boldsymbol{\rho}_i$ in the ring frame, with unit direction $\mathbf{q}_i$ and tension λ_i. The carrier positions follow kinematically, $\mathbf{p}_i = \mathbf{p}_L + R(\mathbf{q}_L)\boldsymbol{\rho}_i - l_i\mathbf{q}_i$, and the thrust each must produce follows from its link's acceleration and tension. At every tick the planner solves

$$\begin{aligned}\min_{\mathbf{x}_{0:N_h},\,\mathbf{u}_{0:N_h-1}} \; & \sum_{k=0}^{N_h-1} \|\mathbf{y}_k - \mathbf{y}_k^{\mathrm{ref}}\|_W^2 + \|\mathbf{y}_{N_h} - \mathbf{y}_{N_h}^{\mathrm{ref}}\|_{W_{N_h}}^2 \\ \text{s.t. } & \mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k; \{\boldsymbol{\rho}_i, l_i\}, \mathbf{d}), \quad C_0\mathbf{x}_0 = C_0\hat{\mathbf{x}}(t), \\ & \lambda_{\min} \le \lambda_{i,k} \le \lambda_{\max}, \quad f_{\min} \le \|\mathbf{f}_{i,k}\| \le f_{\max},\end{aligned} \tag{1}$$

where the output $\mathbf{y}$ stacks the ring's position, velocity, attitude error and rate, the link directions, tensions and rates, and the inputs; $\mathbf{f}_{i,k}$ is carrier i's planned thrust, its bounds softened with slack. The horizon has $N_h = 20$ nodes over 2 s, so a node is the 100 ms planner tick. The constraint on node 0 pins only what motion capture measures, the ring's pose and twist and the link directions; the tensions and link rates are left free and warm-started from the previous solution. A lumped force $\mathbf{d}$ on the ring, formed from the integral of its position error in steady flight, removes steady offsets through the model [28], [29]. The references follow from the ring's trajectory by flatness, with the nominal formation tilted along the effective gravity and the tensions scaled with it.

Problem (1) is solved with acados [30] in a real-time iteration scheme [31]: a fixed number of Gauss–Newton SQP iterations per tick (one unless stated; Section IV-D), warm-started, with HPIPM [32] solving the QP. On the rig no new iteration starts after 60 ms of the tick, and a plan not ready by then is replaced by the previous one shifted by a node.

Each carrier's tracker is an MPC on a 17-state model in which the four stick commands are states and their rates the inputs, with the flight controller's rate loop modelled as first order and the identified affine thrust map. It solves every 20 ms over the same 2 s horizon, following the planner's position, velocity and thrust direction. The link's pull enters its model as a known parameter along the horizon: the planned pull, shifted by the difference between the pull measured from motion capture and the planned one at the first node, so that the tracker anticipates the link's force rather than reacting to it.

B. Reconfiguration: release and drop

Two properties of (1) make an in-flight change of team possible. The plate points $\boldsymbol{\rho}_i$ and link lengths l_i are runtime parameters, so only the team size fixes the problem's dimensions, and one solver per size, built in advance and primed on the measured state, covers any layout. And the survivors keep

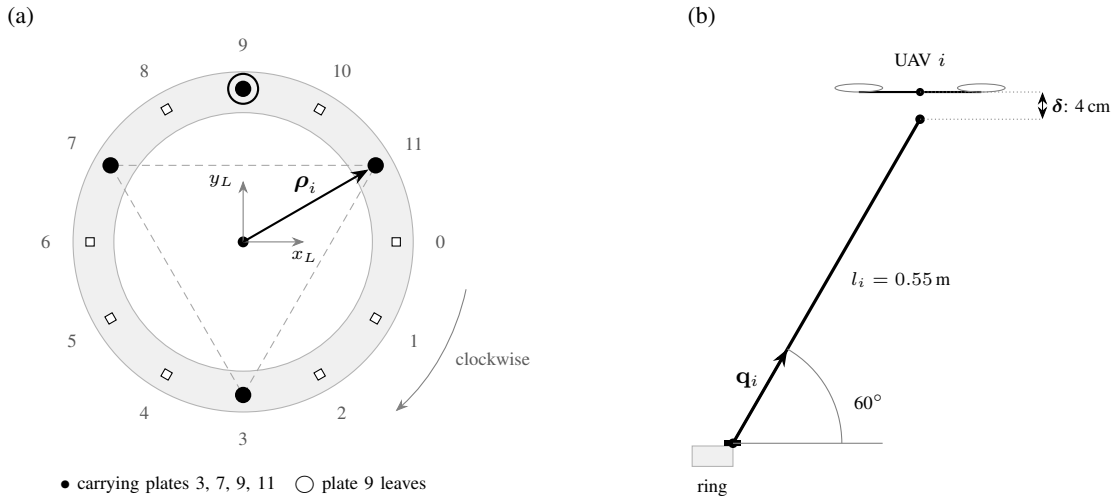


Fig. 2. Payload and rod geometry, to scale. (a) Top view of the 0.86 kg ring with its twelve magnet plates at a radius of 0.225 m, numbered clockwise from the x_L axis seen from above, so that plate k sits $30k$ degrees clockwise from x_L. The four-UAV layout uses plates 3, 7, 9 and 11; when the UAV on plate 9 leaves, the three on plates 3, 7 and 11 form an even triangle (dashed), under which the ring hangs level. (b) Side view of one rod at the planned elevation of 60°: the rod direction $\mathbf{q}_i$ points from the plate to the UAV, and the rod pivots $\boldsymbol{\delta}$ = 4 cm below the UAV's centre. Rig dimensions (0.55 m rods); the twin results use 1 m cables.

their own plates: the nominal tensions come from a moment balance at the actual plate points, so the ring's reference stays level without re-spacing the team.

Detection. A release the planner is told of is a DETACH command, which switches the leaver's magnet off and resizes on the same tick. An unannounced release is detected from the link alone. A rigid link on its plate cannot measure longer than itself, so when the leaver's link, measured by motion capture between the carrier's pivot and its plate point, reads more than 6 cm over its length for two consecutive ticks, the planner removes that carrier. No force sensor and no message from the leaver are needed.

Resize. At the event the planner swaps to the solver for $N-1$ carriers, sets the survivors' plate points and link lengths, and warm-starts it from the measured state with the tensions and link directions free at node 0. The first solve redistributes the load; there is no separate hand-off phase. With three survivors the vertical shares are fixed by statics: each is the ring's weight times the barycentric coordinate of its centre in the survivors' triangle, and a level ring exists only while the centre lies inside the polygon of the plates [3]. For plates on a circle of radius r, with $g_{\max}$ the survivors' largest angular gap, the support margin

$$m = r\cos(g_{\max}/2) \tag{2}$$

is the distance from the centre to the nearest edge. Losing one of four evenly spaced carriers puts the centre on an edge, so the four-carrier layout uses plates 3, 7, 9 and 11, where losing plate 9 leaves an even triangle with a 112 mm margin.

Drop rule. A loss that leaves fewer survivors than the configured minimum (three by default), or a ring tilt above 30°, drops the ring: every magnet is switched off and every carrier lands. Release chains that continue below three carriers fly on while the survivors' largest gap is at most 200°, which keeps a T and an opposite pair. The freed carrier steps out radially, clear of the ring, and lands and disarms alone while the survivors carry on.

C. Rejoin of a departed carrier

A departed carrier can return to its own plate. At its release the planner records its plate point and the link's measured length. *Staging:* the carrier is held on its plate's bearing 0.3 m beyond the formation's reach, the point moving and turning with the planned ring and kept inside the position limit. *Approach:* on the rejoin command, given only while the ring holds its path, the carrier descends at 0.1 m/s so that its link's lower end lands on the plate point of the measured ring, with a feedback loop on the measured end correcting the end's height. *Weld:* the magnet is switched on once the end has stayed within 1.5 cm of its plate point on every sample over 0.3 s (at least eight samples) at under 2 cm/s; the joint must then report attached, or the carrier is let go and backs out to staging. *Fold-in:* the planner grows to $N + 1$ with the plate point and the link re-measured from pivot to plate; the rejoiner is seeded at 0.1 N and every tension is blended to the balanced split over 10 s, with its link feedforward ramped over 1.5 s and the ring's path held throughout, as a dock blends thrust across a switch [20]. Through the blend the rejoiner's height reference is capped at the weld height, so that it cannot lift its plate and tilt the ring before it carries its share, and it counts as absent for the drop rule. LAND or a drop sends every staged or rejoining carrier up, out along its bearing and down alone.

D. Attach of a newcomer

A newcomer is flown by another team's controller down to contact; its magnet catches only there. Our side admits it from motion capture alone, with its magnet tip expressed in the plate's frame. A *contact* is the tip within 24 mm of the plate point and close to the plate's face, nearly still for 0.3 s.

TABLE I
REACTION TO A LOST CARRIER, FOUR CARRIERS IN HOVER. TIMES FROM THE RELEASE (OR THE COMMAND). RIG: 0.55 M RODS; TWIN: 1 M CABLES.

Case	Detected (s)	Resized (s)	Tilt peak (°)	Back in band (s)
Rig, release (Fig. 3)	0.42	0.51	6.5	1.8
Rig, all three releases	0.33–0.42	[to measure]	6.5–17.8	1.8–3.8
Rig, announced detach	–	0	15.4	1.95
Twin, release	–	0.37	5.9	3.4

A contact is only a candidate, since a tip resting on a plate and a caught magnet look alike, so the catch is *confirmed* by co-motion: after the newcomer's small lift, the tip has stayed within 6 mm of its contact point while the plate moved with it, on a taut tether. Only then do our trackers take the newcomer over and the planner grows to $N + 1$ with the recorded plate point, the newcomer seeded at 0.1 N and blended as in Section III-C. A newcomer not confirmed in time is taken back: our tracker latches onto its measured state, lifts it clear and lands it alone.

IV. RESULTS

We test our planner on the rig and in its simulated twin. The rig carries the 0.86 kg ring with magnet plates every 30° on its rim; the twin models the same ring, the same quadrotors and the rig's thrust map and motion-capture filter, but its cable links are rigid. Unless stated otherwise the twin flies 1 m cables with the ring at 1.0 m. Bars were written before each set of runs, and a run is scored against the bars of its set only. A ring is *in band* when its tilt is within [band definition] of level; *tilt rise* is the peak tilt after an event less the tilt just before it, while the drop rule acts on the tilt itself.

A. Release and rejoin

a) An unannounced release on the rig: Four carriers hover the ring on the rig, and one of them lets go of its plate without telling the planner. Fig. 3 shows the reaction of Section III-B. The planner sees the released cable stretch past its threshold 0.42 s after the release and resizes its problem to three carriers 0.09 s later, each survivor then planned at about 3.2–3.3 N. The ring's tilt peaks at 6.5° and is back in its band 1.8 s after the release (Table I). Over all three rig releases, detection took 0.33–0.42 s and the ring was back in band within 1.8–3.8 s, with tilt peaks of 6.5–17.8°; the slowest, with the largest tilt, left a 150° gap between survivors. Three releases are too few for statistics; they show the reaction, not its spread. An announced detach, by contrast, resizes at the command but tilted the ring further, to 15.4°. In the twin, the same release on 1 m cables resized after 0.37 s with a 5.9° tilt peak but took 3.4 s to return to band, so the rig and twin rows differ in link length as well as in plant and are not a like-for-like check.

b) A departed carrier rejoins (twin): A released carrier is not landed: it holds a staging point beyond its own plate and, on the operator's command, flies back, is re-attached and is folded into the team (Section III-C). This is a twin result

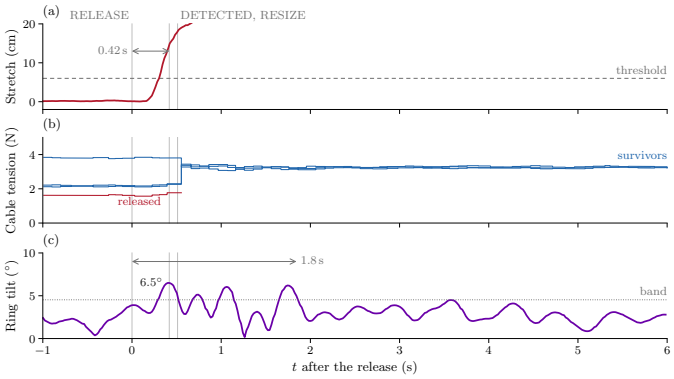


Fig. 3. An unannounced release on the rig: one of four carriers (red) lets go of its plate in hover. The planner detects the loose cable from its stretch, resizes to the three survivors (blue), and the ring's tilt (black) returns to its band.

TABLE II
A DEPARTED CARRIER REJOINS ITS OWN PLATE (TWIN, HOVER, 1 M CABLES), FINAL CODE TREE. BARS, FROZEN BEFORE THE RUNS, IN BRACKETS. LEANING: A STEADY TORQUE LEANS THE T'S PAIR ABOUT 3°. RANGES OVER n RUNS.

Rejoin	n	On plate after (s) [30]	Weld after (s) [0.5]	Fold-in tilt (°) [5]	End from plate (cm) [2]
Square	4	23.9–24.9	0.029–0.036	1.27–1.55	1.3–1.6
T	3	16.6–17.7	0.012–0.021	1.45–3.02	1.3–1.5
T, leaning toward	2	15.6–16.5	0.015–0.018	3.12–**5.28**	1.3–1.5
T, leaning away	2	17.7–18.6	0.015–0.018	2.91–3.37	1.4–1.5

only, since the twin's joint welds the cable wherever its end lies, while the rig's magnet catches only on contact. A rejoin is scored in stages, each against a bar frozen before the runs: the cable's end on its plate within 30 s; the joint attached within 0.5 s of that; then the fold-in, with the ring's tilt at most 5° from the weld to 15 s after it, the end within 2 cm of its plate through the 10 s blend, and the rejoiner's planned pull at the blend's end within 10 % of its share. Table II gives the results onto a square (four carriers back from three) and onto a T (three back from an opposite pair); Fig. 4 shows one rejoin onto the T.

Every rejoin welded within 1.5 cm of its plate point, and 10 of the 11 fold-ins tilted the ring by 1.3–3.4°. With a steady torque leaning the T's pair about 3°, the rejoin passed 3 of 4 runs: in one run leaning toward the rejoin plate the ring tipped to 5.28° for 0.19 s, missing the bar by 0.28°. There the rejoiner's height reference, held at or below its weld height through the blend to stop its rigid link lifting the T's free side, dipped 1.1 cm at the fold-in. The hold patches a twin artefact, a rigid link, which the rig's slack tethers do not share. Two neighbouring checks also passed: a LAND during the rejoin's descent sends the rejoiner up, out and down alone, 1.245 m from the ring's centre (bar 1.2 m), and a carrier staged through a figure-eight lap of the T stays inside the position limit, with its attach command refused while the path moves.

c) Eight to four and back (twin): Release and rejoin compose into a chain. Eight carriers at 45° on 1 m cables lift a 0.9 kg ring of radius 0.5 m from the floor; every other carrier

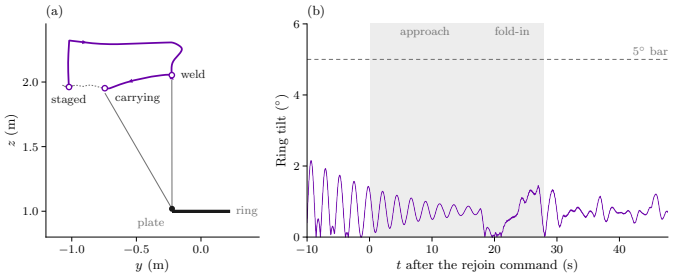


Fig. 4. A carrier released from the T flies back to its own plate in the twin, re-welds 17.8 s after the rejoin command and folds back in. (a) The rejoiner's path, side view, with its cable at the weld and folded in; (b) the ring's tilt, peaking at 1.45° in the fold-in, against the 5° bar.

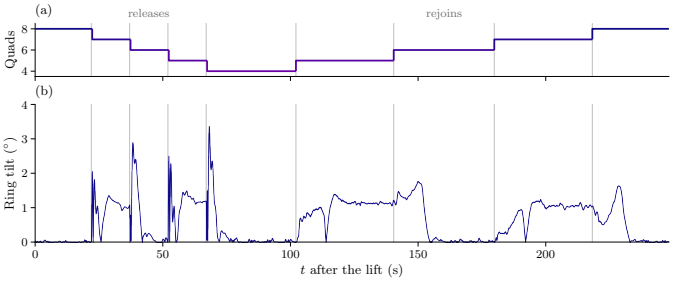


Fig. 5. Eight carriers take the 0.9 kg, 0.5 m ring through four unannounced releases down to four and four rejoins back to eight (twin, not real time; an earlier exploratory run at one iteration a tick, the claim runs at five described in the text). (a) The team size the planner flies; (b) the ring's tilt, peaking at 3.35°.

is released unannounced, one at a time and 15 s apart, down to a square of four, and the four freed carriers then rejoin one at a time back to eight (Fig. 5). In both claim runs, at five solver iterations per tick, no drop or abort fired; each release raised the ring's tilt by at most 2.9° with the planner resized within 0.4 s, and each rejoin's planner grew within 16.2–19.4 s of its start. Back at eight, scored from 2 s after the last rejoin's blend ends, the ring's tilt held its bar (0.95–0.99°) but its position error, 6.4 mm (rms) in both runs, missed its bar of 5.7 mm: the last fold-in leaves the ring 36 mm off, and it returns over about 5 s. The world runs at about a quarter of real time here, so these runs are about the plan and the physics, not about timing (Section IV-D).

d) A newcomer attaches on the rig: [The rig attach of a newcomer on 1 m cables (three carriers to four with the basket): flies 29 Oct.]

B. What sets the release transient

The rig's slowest release left the largest gap between survivors (Section IV-A). To isolate that variable, the twin flies the same unannounced release from four carriers in hover, with the carriers placed so that the three survivors leave a largest gap of 120 to 180° around the ring. Everything else is held fixed. We report the rise of the ring's tilt from the release to 15 s after it, cut at the moment the ring is dropped or a fault stops the flight. One arm at 150° feeds the survivors' trackers the planned cable pull instead of the measured one (Section III-A). The 0.55 m column is one run a cell.

TABLE III
RISE OF THE RING'S TILT AFTER AN UNANNOUNCED RELEASE FROM FOUR CARRIERS IN HOVER (TWIN), AGAINST THE SURVIVORS' LARGEST GAP. REPEATS AS A / B; ≥: CUT AT THE DROP OR THE FAULT. MEASURED CABLE PULL UNLESS MARKED.

Largest gap (°)	Tilt rise, 1 m (°)	Tilt rise, 0.55 m (°)	Outcome, 1 m
120	6.90 / 7.45 / 7.50	6.40	held
135	10.42 / 11.02	8.44	held
150	18.17 / 17.58	16.82	held
150, planned pull	29.62 / 27.10	≥29.54	held (0.55 m: dropped)
165	≥30.49 / ≥29.84	28.02	dropped (0.55 m: held)
180	≥62.46 / ≥61.91	≥61.22	tilt fault

Table III gives the results. The tilt rise is steep in the gap: from 120 to 150° it more than doubles, at 165° it crosses the 30° line at which the ring is dropped and every carrier lands (2 of 2), and at 180°, with the survivors on one half of the rim, it passes 60° and the fleet is stopped by the tilt fault. Every neighbouring pair of gaps is separated by more than the spread of its repeats at 1 m. The longer cable changes the boundary only slightly: at 0.55 m the 165° gap rose just under the drop line and the ring was held (one run). At 150° the measured pull holds the rise to about half of the rise with the planned pull at 0.55 m and to about two thirds at 1 m, where the planned arm stays just under the drop line instead of crossing it.

The controller cannot undo a bad geometry: once the survivors leave a wide gap, the ring has lost its support on that side, and no redistribution of pull among them restores it within the drop line. The leaving order is therefore part of the design. An order that keeps the survivors' gaps small keeps the transient small; opposite pairs are the simplest such order. The $8 \to 4 \to 8$ chain of Fig. 5 releases its carriers in opposite pairs, so each release leaves gaps of at most 90°.

C. How far off its plate a newcomer can catch

A magnet catches wherever its tip is when it fires, which need not be the centre of its plate. The planner of Section III-D therefore folds the newcomer in at the *measured* catch point, the tip's position in the ring's frame at the catch. We test this choice against folding in at the slot assumed, the plate's nominal point, in the twin: three carriers hover the ring on 1 m cables, the newcomer's catch point is moved off its plate by a set distance d, either along the rim toward the neighbouring carrier or radially inward, and the fleet then flies a figure-eight. Each cell flies twice. A run completes if the newcomer is caught and folded in, the ring's tilt peak in the fold-in stays under 5°, and the fleet finishes its path. Up to about 2.5 cm the offset is a real catch error, since a real magnet catches only on its 60 mm plate; beyond that the sweep measures the fold-in's tolerance to error in the attach-point model, not how far off a real magnet can catch.

Radially inward the measured catch point holds to the largest offset flown, 20 cm, with a fold-in tilt peak of at most 2.63° (Fig. 6, Table IV). Assuming the slot passes the 5° bar from 8 cm: the planner's balance ignores the moment of the offset, so the ring hangs off its reference and the newcomer

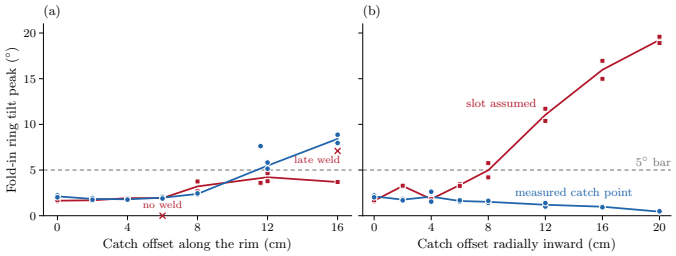


Fig. 6. The ring's tilt peak in the fold-in against the catch offset, folding in at the measured catch point and at the slot assumed, (a) along the rim and (b) radially inward. One marker per run, the line through the cell means; crosses mark the slot arm's two weld-trigger failures. The bar is 5°. Past about 2.5 cm the offset is model error rather than a real catch.

TABLE IV
LARGEST CATCH OFFSET COMPLETED WITHIN THE 5° FOLD-IN BAR (TWO RUNS A CELL, TWIN, 1 M CABLES).

Fold-in point	Radially inward	Along the rim
Measured catch point	20 cm	8 cm
Slot assumed	6 cm	4 cm

nears its throttle limit, and the tilt grows with the offset to 15–17° at 16 cm and 19–20° at 20 cm, where the measured arm peaks at 0.4–0.5°. Here the mechanism is shown: the slot arm's tilt grows with the model error, the measured arm's does not.

Along the rim the order reverses past 8 cm. The measured catch point completes to 8 cm and fails the fold-in bar at 11.6 cm, the next plate's centre, with tilt peaks of 5.1–8.9° from there to 16 cm. The slot arm folds in more gently along the rim, peaking at 3.5–4.7° over the same offsets; its own limit, 4 cm, is set by the twin's weld trigger, which once failed to fire at 6 cm whatever the arm. Along the rim the mechanism is therefore not shown. Separation from the neighbouring carrier is reported apart from these limits: folding in at the measured point moves the newcomer toward that carrier, to 0.22–0.23 m at 16 cm, inside the 0.30 m line, while the slot arm keeps at least 0.54 m and every radial run at least 0.64 m.

D. Scalability

Our planner solves the OCP (1) with a fixed number of sequential-quadratic-programming (SQP) iterations per 100 ms tick, warm-started from the previous plan. Fewer iterations shorten the tick but leave the plan short of convergence. We measure what that costs against the fleet size in two ways. In closed loop, the four-carrier twin flew a hover, the figure-eight at 0.2 m/s and an unannounced release, each from a floor start, at one iteration and at five. Offline, each plan at one and two iterations was compared with the plan converged from the same iterate. The bars of both checks were fixed before the runs: the dynamics residual of the plan, the error of the references the trackers fly at the next node and the fifth, the error of the thrust feed-forward, and the cost's ratio to the converged plan's.

In closed loop at four carriers, one iteration a tick flew as well as five in every cell. The ring's path error and tilt stayed inside the bars set from the five-iteration runs through

TABLE V
PLAN QUALITY AT EIGHT CARRIERS AGAINST SQP ITERATIONS PER TICK: THE FIGURE-EIGHT AT 0.2 M/S, EACH WARM START 1 CM AND 10 CM/S OFF THE PLAN, COMPARED WITH THE PLAN CONVERGED FROM THE SAME ITERATE; 95TH PERCENTILES OVER 200 TICKS. BOLD: OVER THE BAR. NO SOLVE FAILED.

Measure	Bar	5	2	1
Cost ratio, median / 95th pct.	1.10 / 1.50	1.000 / 1.001	1.003 / 1.010	**miss**[a]
Dynamics residual	10^{-3}	3.2×10^{-7}	8.8×10^{-5}	$\mathbf{9.3\times10^{-3}}$
Reference, node 1 / 5 (mm)	5 / 10	0.9 / 0.9	3.2 / 5.1	**7.3 / 14.5**
Feed-forward (m/s^2)	0.3	0.12	**0.316**	**0.60**

[a] 170 of 200 ticks over the residual bar, each counted as a miss.

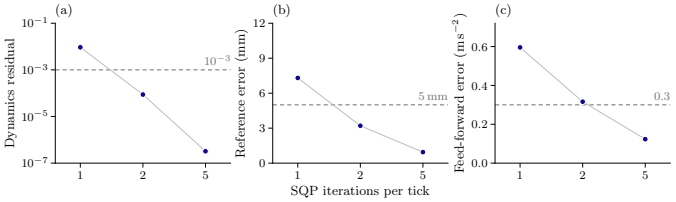


Fig. 7. Plan quality at eight carriers against SQP iterations per tick, relative to the converged plan: the dynamics residual, the reference error at the next node and the thrust feed-forward error (95th percentiles), with the bars fixed before the runs. One iteration misses every bar, two miss only the feed-forward bar, five pass.

the lift, the path and the release, and no run dropped, aborted or failed a solve. The plans were not converged, however: their dynamics residual peaked at the release, under the closed loop's limit but over the offline bar.

Offline, the answer depends on the fleet. Replayed tick by tick, nine recorded flights with two to four carriers, the rig's among them, converge by every bar at two iterations; at one iteration every rig flight breaks the residual bar. At eight carriers on the figure-eight, with each warm start moved off the plan by about the rig's ring error, one iteration misses every bar, two miss only the feed-forward bar (0.316 against 0.3 m/s^2), and five pass all of them (Table V, Fig. 7). The iteration count must therefore grow with the fleet: two suffice up to four carriers, while eight need five.

What this means for real time is set by the slowest tick. On the rig laptop, a bench calibrated against the rig's own flights projects, as an upper bound, the largest team whose every tick runs its full iterations within the 100 ms period: three carriers at five iterations and four at two (the bench with the reference speed-ups on, scaled by the rig's measured in-flight slowdown of 3.5 from three-carrier flights). This is not a flight limit: the rig has flown four carriers at five iterations under its 60 ms solve budget, which cuts a late tick's iterations short. At eight carriers the plan must be read at five iterations, which the rig laptop cannot run in real time; the eight-carrier results are therefore the twin's, with its clock slowed, and say nothing about real time. In the twin, at five iterations and two runs each, eight carriers held a hover with the ring within 1.3 mm (rms) and 0.06° of tilt, and carried the ring $8 \to 4 \to 8$ (Fig. 5) with no drop: each release raised the ring's tilt by at most 2.9°, and back at eight the tilt held its bar while the position error, 6.4 mm, missed its bar of 5.7 mm (Section IV-A's chain). These runs flew at about a quarter of real time, with late ticks

under 0.3 %.

V. Discussion

a) Rigid links in the twin, slack tethers on the rig: The twin models every cable as a rigid link, which can push as well as pull and welds wherever its end lies. This flatters the rejoin and also creates its own failure: on a T, the rejoiner's rigid link set the ring's pitch, so a rising height reference lifted the ring's free side, and the height hold that cured it is a patch for the twin's links. On the rig the tethers go slack, so that lift cannot arise, but the attach is harder: the magnet catches only on contact, which is why the rig's attach is confirmed from motion capture before the newcomer is folded in. The rig's release reaction was measured on the rig itself; the twin's release on 1 m cables gave a similar tilt peak but a slower return to band, and the rig flew 0.55 m rods, so the two are not a like-for-like validation.

b) Timing: Every reconfiguration in Section IV is paced by detection and blending rather than by the solver: a lost carrier is seen from its cable's stretch in 0.33–0.42 s and the problem is resized within about 0.1 s more, and a rejoin's fold-in is blended over 10 s. The solver sets the size of the team instead. At four carriers one SQP iteration per tick flies as well as five in the twin's closed loop, but its plans are not converged on the rig's flights, and at eight carriers only five iterations pass every bar (Section IV-D). The eight-carrier chain ran slower than real time, so it shows that the plan and the physics compose, not that the rig's laptop can fly it.

c) Geometry sets how hard a release hits: The release transient is set less by the controller than by the team that remains. The ring's tilt after a release grows steeply with the largest gap the survivors leave around the rim, and on 1 m cables a 165° gap already crosses the 30° drop line (Section IV-B). The planner can shorten the transient, most of all when it plans from the measured cable pull rather than the planned one, but it cannot remove the torque that an uneven ring of survivors applies. A team that must shed carriers should therefore leave in an order that keeps the survivors' gaps small, and a drop rule written in terms of that gap, rather than of the number of carriers left, states what the ring can bear. On the attach side, folding in at the measured catch point makes the twin's fold-in tolerant of catch-point error, 16 cm radially and 6 cm along the rim (Section IV-C); past about 2.5 cm this is tolerance to model error, not a catch a real magnet could make.

d) Limitations: The rig results cover the release only, three runs on 0.55 m rods; the rejoin, the attach-offset sweep, the gap sweep and the eight-carrier chain are twin results, two runs a cell or fewer, on rigid links that weld wherever the end lies. No reconfiguration is compared with a baseline that does not resize the problem. The eight-carrier chain ran at about a quarter of real time, at the five iterations the offline check asks for at that size, and its return to eight missed the position bar by 0.7 mm; and along the rim the measured catch point's advantage over the slot assumed is not shown.

e) Next on the rig: The next rig flights test the attach of a newcomer on 1 m cables, three carriers to four with a basket, with the attach confirmed by motion capture before the fold-in; the four-carrier hover and paths on 1 m cables; and the planner's tick in flight with the speed-ups, which sets the real-time limit. The rejoin of a departed carrier, which in the twin depends on the joint's weld, waits for a rig attach that has been shown repeatable.

VI. Conclusion

We presented a planner that lets a team of quadrotors carry a cable-suspended ring through changes of its own membership in flight. On the rig, an unannounced release of one of four carriers was detected in 0.42 s, the team was resized 0.09 s later, and the ring tilted at most 6.5° before returning to its band in 1.8 s. In the twin, a departed carrier flew back to its own plate, welded within 1.5 cm and was folded in with the ring tilting 1.3–3.4° in 10 of 11 runs, and in one exploratory run eight carriers took a larger ring down to four and back with a tilt peak of 3.35°. In offline replays of nine flights, two SQP iterations per tick converge for every team of up to four, and the iterations needed grow with the size of the team (Section IV-D). How hard a release hits is set by the survivors' geometry, the largest gap they leave, and folding a newcomer in at its measured catch point tolerates large errors in that point. The rig attach of a newcomer on 1 m cables is the next step [result from 29 Oct], together with the real-time limit of the planner's tick on the rig [real-time limit].

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